

Open Problem Answers

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1. Pre-Specified Open Problems

The assignment specifies three open problems requiring original thought. The P1-P4 scoring reference

2. P1-P4 Scoring Conflict

2.1 The Conflict

The assignment specifies:

- P1-P4 references "three answers / three pages"
- But lists P1, P2, P3, P4 = four problems

This creates ambiguity:

- Three answers in three pages?
- Four answers in some allocation?
- Which three problems are primary?

2.2 Recommended Interpretation

Given the explicit "three answers / three pages" limit, the recommended interpretation is:

****Select three of the four problems (P1-P4) to answer in three pages total.****

1. ****P1**** (What would you do differently?) - Include
2. ****P2**** (What is still uncertain?) - Include
3. ****P3**** (What would you investigate next?) - Include
4. ****P4**** (What is the most surprising finding?) - Exclude OR combine with P2

This interpretation respects both constraints: three answers fitting three pages while covering the

3. Open Problem Answers

3.1 P1: What would you do differently?

****Answer:****

Given the identification failures in this assessment, I would fundamentally change the study design:

1. ****Require independent service activation records**** before any analysis. The treated services must
2. ****Separate the scheduling policy change**** from the system implementation. Either delay one or imp

3. **Establish ground truth through adjudication** before analysis. Two unadjudicated reviewers are
4. **Implement the study as a proper target trial** with randomization or instrumental variable, not
5. **Build authorization into data collection**, not post-hoc. The authorization model should be enf

3.2 P2: What is still uncertain?

Answer:

Many critical uncertainties remain:

1. **Treatment assignment:** Which services actually switched on? The all-Site A proxy is a strong a
2. **Endpoint validity:** Can `cancel\_rate` be mapped to day-of-surgery cancellation? This requires
3. **Label representativeness:** The 500 labeled pairs are not random. What fraction of total recomm
4. **Reviewer correctness:** Two reviewers disagree on some cases. Without adjudication, we don't kn
5. **Generalizability:** The three-month post-switch window may not represent steady-state behavior.
6. **Interference:** The 8-component structure may not capture all clinician cross-coverage. The int

3.3 P3: What would you investigate next?

Answer:

Given the findings, the next investigations would be:

1. **Obtain the service activation contract:** What services were actually selected? This is the cri
2. **Validate the cancellation endpoint:** Work with clinical stakeholders to define day-of-surgery
3. **Adjudicate the label disagreements:** Get a third clinician to resolve the 120 reviewed pairs.
4. **Extend the follow-up period:** Three months is insufficient for understanding long-term effects
5. **Test the authorization model:** Run the analytics assistant against the full sealed question se
6. **Simulate the sequential operating characteristics:** The monitoring design specifies but does n

4. Additional Open Questions

4.1 Why did the analyst's query have so many defects?

The analyst SQL had mechanical issues (join loss, duplication) that suggest:

- Inadequate testing/validation
- Lack of service-level understanding
- Pressure to produce results

4.2 What caused the scheduling policy co-intervention?

The policy change occurred at the same time as the system switch. Understanding the policy would hel

4.3 Why is there no ground truth?

The assessment lacked:

- Adjudication of reviewer disagreements
- Independent validation of LLM judge
- Clear endpoint definitions

5. Summary

The open problems reveal fundamental gaps in the assessment:

- Missing treatment identification
- Undefined endpoints
- No ground truth
- Insufficient design

These are not fixable with better analysis?they require redesign of data collection and study implem

*This document addresses the open problems as required. The P1-P4 conflict is flagged for decision: